

Birla Institute of Technology & Science, Pilani, Rajasthan
First Semester 2025-2026
Tutorial-1

Course: ECE/EEE F311 Communication Systems
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Objectives

- Understand the concept of block-diagram approach, transmission bandwidth, and Low pass filtering to input signals.

Notations

$$\text{sinc}(x) = \frac{\sin(x)}{x}.$$

Tutorial Problems

1. A signal $m(t) = 8000\text{sinc}(8000\pi t)$ is passed through a channel with impulse response (i) $h(t) = 9000\text{sinc}(9000\pi t)$ (ii) $h(t) = 4000\text{sinc}(4000\pi t)$. Discuss qualitatively about the output signal for both the cases.
2. The human voice signal typically contains significant frequency components up to approximately 3.3 kHz. To model this behavior mathematically, consider a signal $m(t)$ composed of $N = 30$ harmonically related sinusoidal components with a fundamental frequency of $f_0 = 110$ Hz. The signal is defined as

$$m(t) = \sum_{k=1}^{30} \frac{1}{k} \cos(2\pi k f_0 t).$$

- (a) Compute the average power of the signal $m(t)$.
- (b) Determine a finite-duration speech segment by applying a rectangular window of width $t_0 = 10$ ms to produce the truncated signal

$$m_w(t) = m(t)w(t), \quad \text{where} \quad w(t) = \begin{cases} 1, & 0 \leq t \leq t_0, \\ 0, & \text{otherwise.} \end{cases}$$

Thus, determine the effective bandwidth of the windowed signal $m_w(t)$ by estimating the main lobe width bandwidth of the rectangular window.

- (c) A channel $h(t) = 8000\text{sinc}(8000\pi t)$ is available for transmission. Draw the transmitter block diagram.

3. In digital communication systems, information is transmitted as a stream of binary digits such as 01100011. . . . For physical transmission over a channel, each bit—'1' or '0'—must be represented by a corresponding waveform. Let bit '1' be represented by the signal $m_1(t)$ and bit '0' by $m_0(t)$. Model the digital transmission waveform by appropriately mapping the bit stream into a continuous-time signal. Assume $m_1(t)$ as a rectangular pulse of width T_1 with amplitude $+A$ and $x_0(t)$ as rectangular pulse of width T_0 with amplitude $-A$. Compute the average energy per bit E_b . Repeat the same analysis for the case where $m_1(t) = A\text{sinc}(\pi t/T_1)$ and $m_0(t) = -A\text{sinc}(\pi t/T_0)$.
4. You are transmitting a voice signal $m(t)$ to a friend in Delhi using a landline telephone system. The two telephone systems are interconnected via a twisted pair cable. Assume that the overall communication system can be modeled as a linear time-invariant (LTI) system, where the transmitter has an impulse response denoted by $g_1(t)^{\text{tx}}$, the transmission channel by $h(t)$, and the receiver by $g_2(t)^{\text{rx}}$. Mathematically express the received signal $\hat{m}(t)$ as a convolution of these components. Design an appropriate receiver filter to optimally recover the original message signal $m(t)$, and discuss the practical challenges that may arise during signal detection.